IB AA HL Solver - keying-in guide

KhiCAS on TI-Nspire CX. Jump code = type on the main menu, e.g. 511.

Keys

`x^2`	[x²] (shows **2, fine)
`^ any power`	[^]
`sqrt(`	[ctrl] [x²]
`e^( (exp)`	[e^x] types exp(close with)
`ln(`	[ctrl] [e^x]
`log( base 10`	[ctrl] [10^x] types log10(
`sin( cos( tan(`	[trig] [shift][trig] [ctrl][trig]
`pi`	[pi] key
`| modulus`	[shift] [=]
`i complex`	letter i
`" quote`	[ctrl] [×]
`< >`	[ctrl][-] [shift][+]
`_ (log_3)`	[shift] [-]
`negative`	[(-)] or [-]

Examples

Polynomial **511**

`-2x^4+3x^3-x/2+7`

leading minus: [(-)]

Rational **511**

`(2x^2-3x+1)/((x-1)(x+2)^2)`

bracket the whole top and bottom

Root / power **511**

`x^2sqrt(4-x^2)+(x^2+1)^(1/3)`

cube root = ^(1/3)

Exponential **511**

`3e^(-2x+1)-2^(x+1)+e^-x`

e^(= [e^x]; e^-x needs no brackets

Logarithm **511**

`ln(x^2+1)-2log(x)+ln(x)/ln(3)`

log=base 10; log3(x): ln(x)/ln(3)

Log equation **121**

`log_3(x)+log_3(x-8)=2`

log_b(only in this solver

Trig **511**

`3sin(2x-pi/3)^2+cos(x/2)`

sin^2(u): type sin(u)^2; theta: t

Trig equation **331**

`2sin(2x-pi/3)=1`

then d/r and the interval

Reciprocal trig **511**

`sec(x)^2-cosec(2x)+cot(x)`

type the letters; csc also ok

Inverse trig **511**

`arcsin(x/2)+arctan(3x^2)`

asin(acos(atan(also ok

Modulus **232**

`|2x-3|=|x+1|`

| = [shift][=]; nested: abs(abs(x)-1)

Composite **511**

`ln(sqrt(1+e^(2x)))`

close every bracket

Product/quot. **511**

`x^3e^(-x^2)/(1+x)`

no * needed: 2x(x+1), x^2y

Implicit x,y **514**

`x^2y+y^3=2x+1`

then the point: (1,1)

Diff. eqn **573**

`(x^2+y^2)/(xy)`

type only the right side of dy/dx =

Complex **132**

`1+sqrt(3)i`

then n = 5 for z^5 (De Moivre)

Pitfalls

Don't type	because	type instead
`sinx^2`	means sin(x^2)	`sin(x)^2`
`sin^2(x)`	error	`sin(x)^2`
`1/2x`	means x/2	`1/(2x)`
`theta`	becomes t*h*e*t*a	`t`
`f(x)=x^2`	f(x) breaks it	`x^2`
`||x|-1|`	error	`abs(abs(x)-1)`
`inf, oo`	not supported	`-`
`log(x)`	is base 10	`ln(x) for base e`